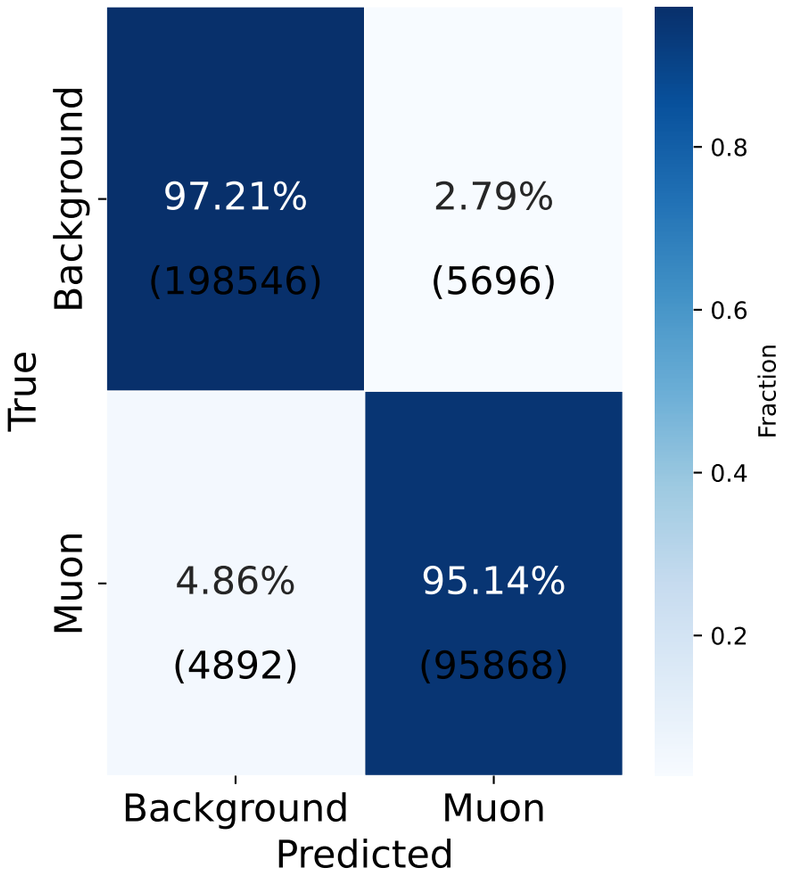
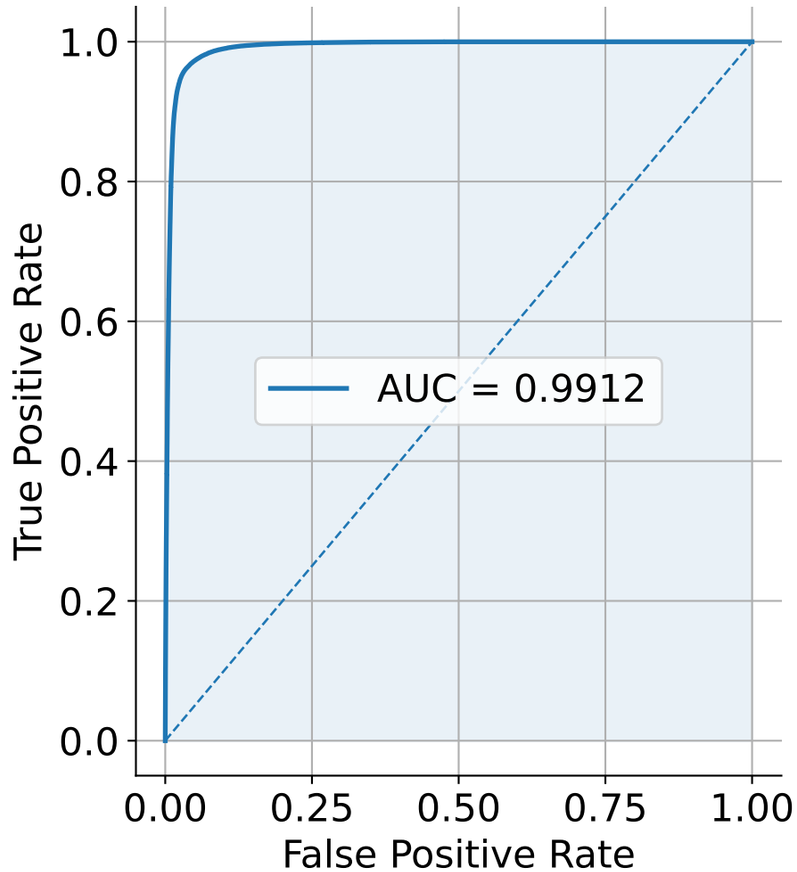
Muon Identification — XGBoost Analysis (hit-based features)
[Train S/B=1:2 | Test S/B=1:2 (realistic, all files)]

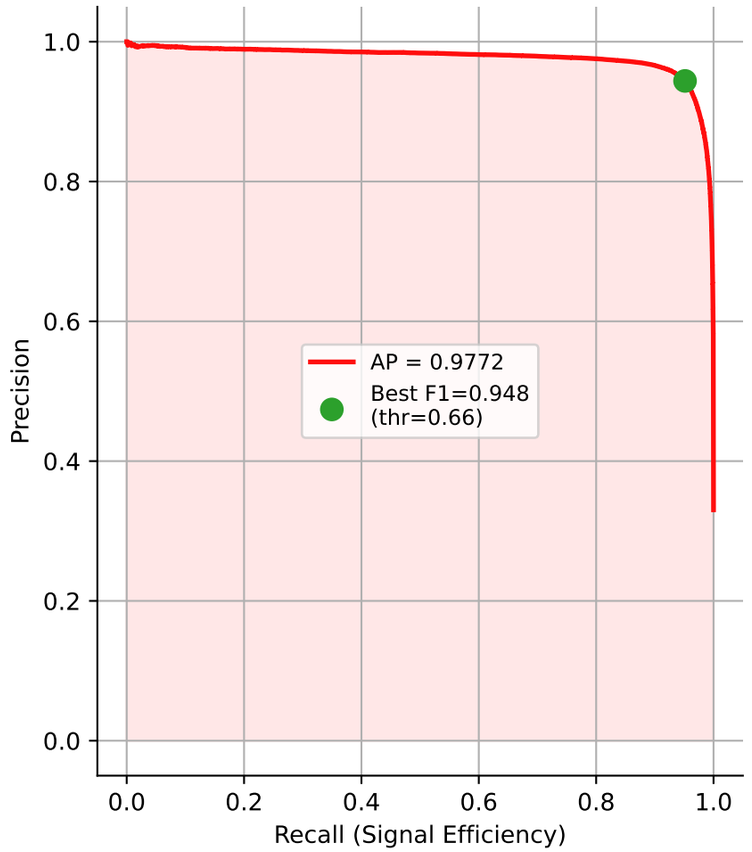
Confusion Matrix @ Max F1-Score (thr=0.663)



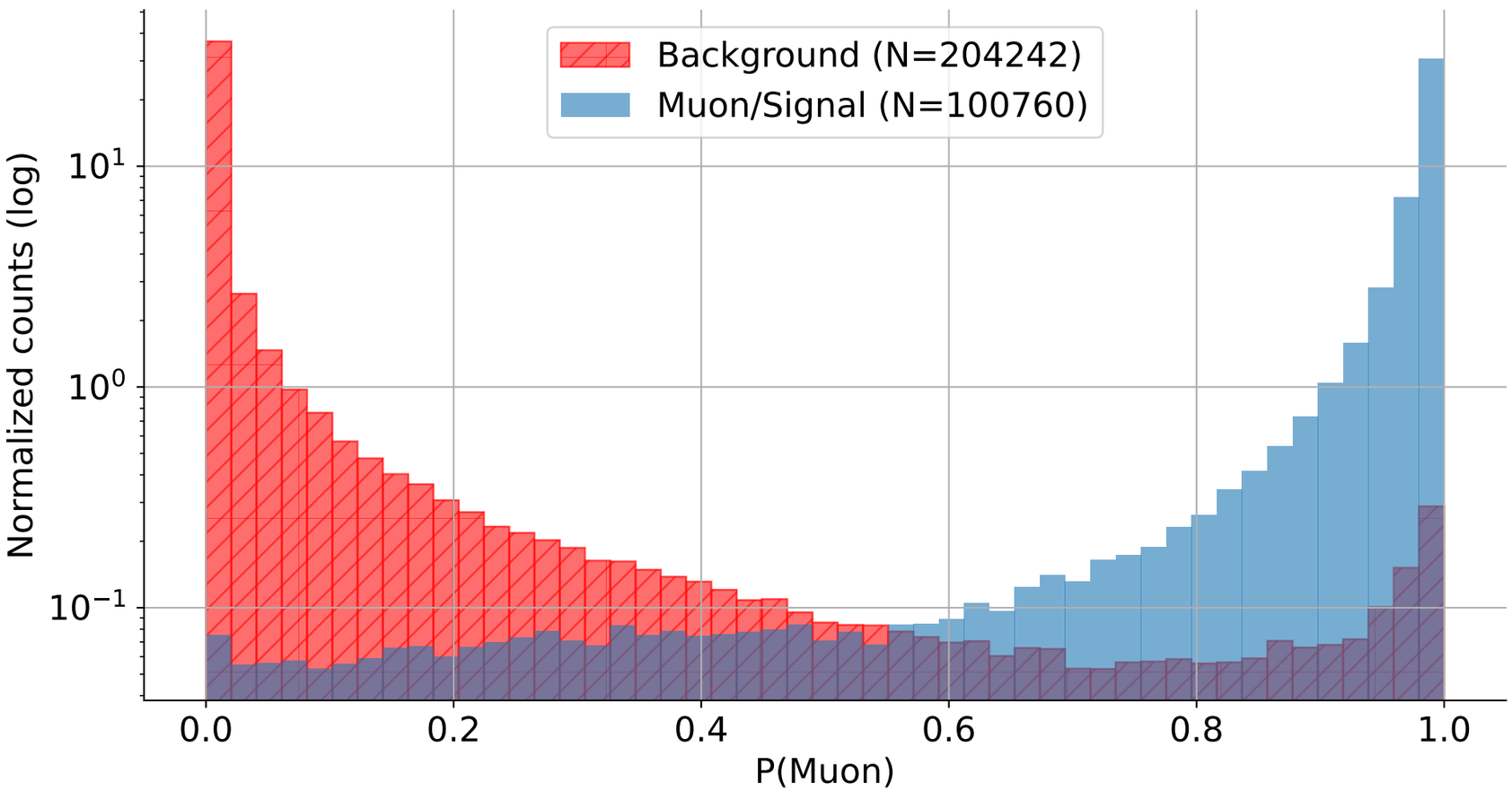
ROC Curve



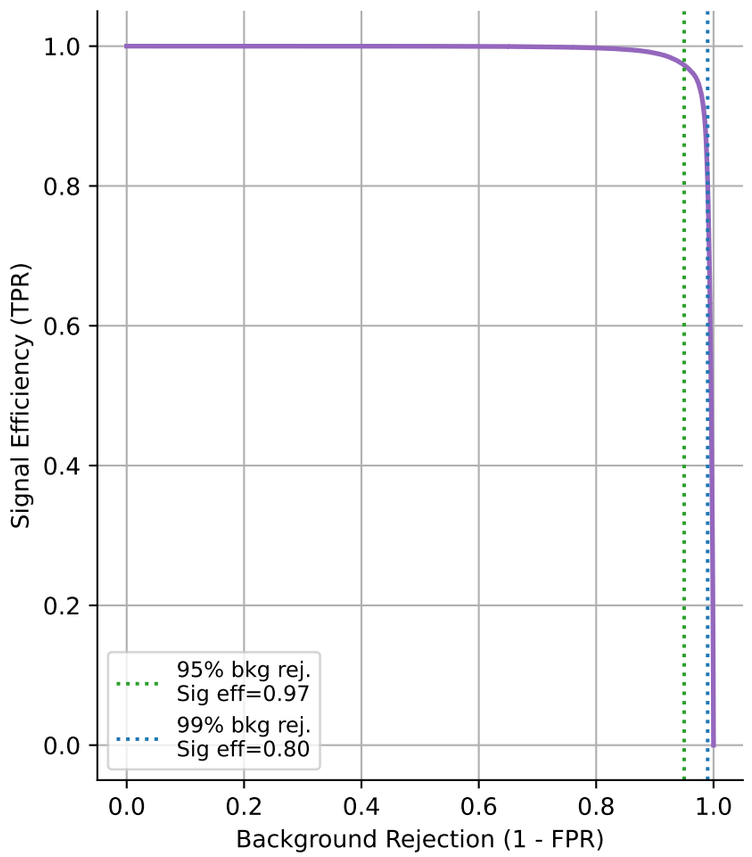
Precision-Recall Curve



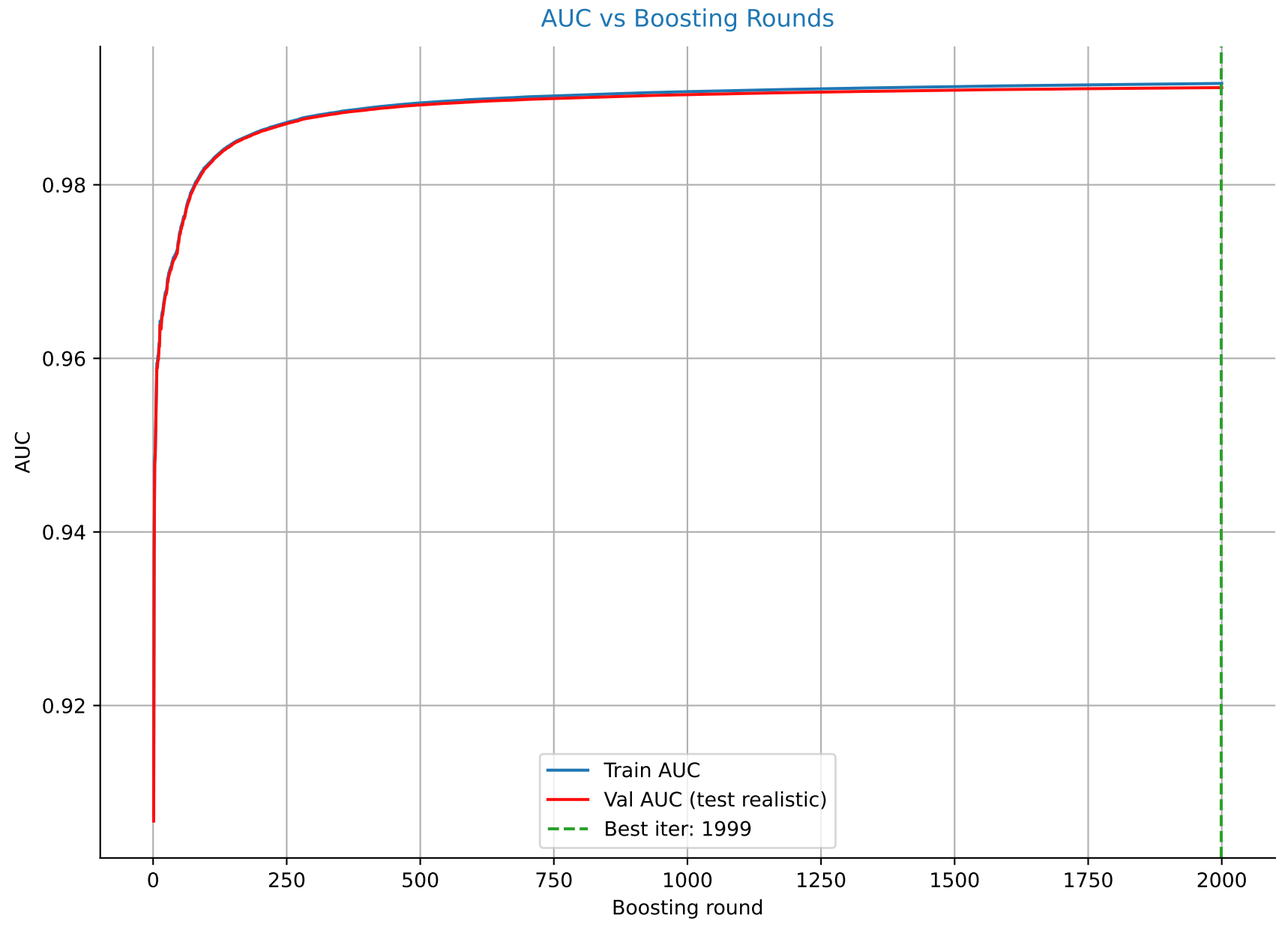
Classifier Response Distribution [Test: realistic S/B]

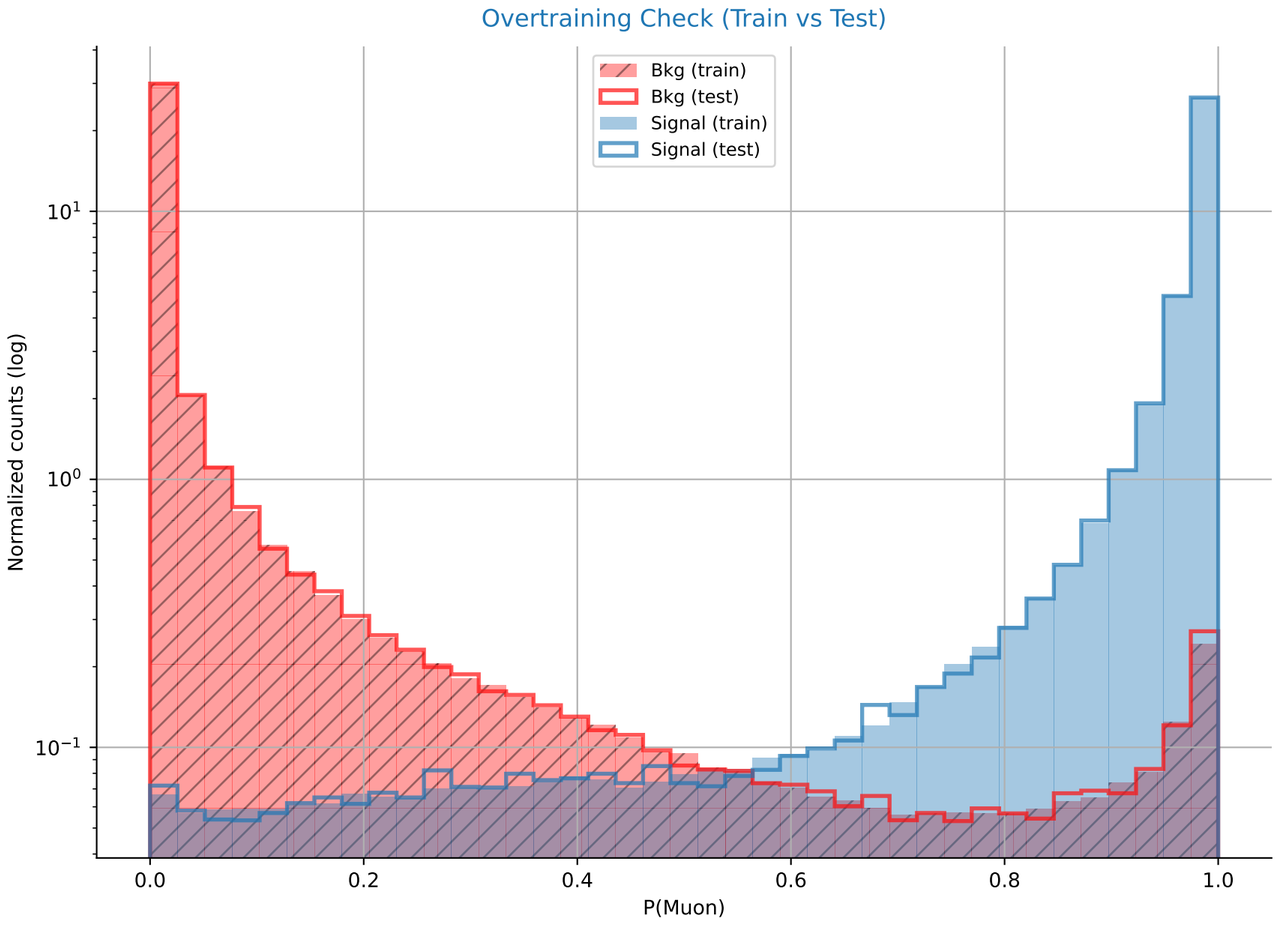


Signal Eff. vs Background Rejection

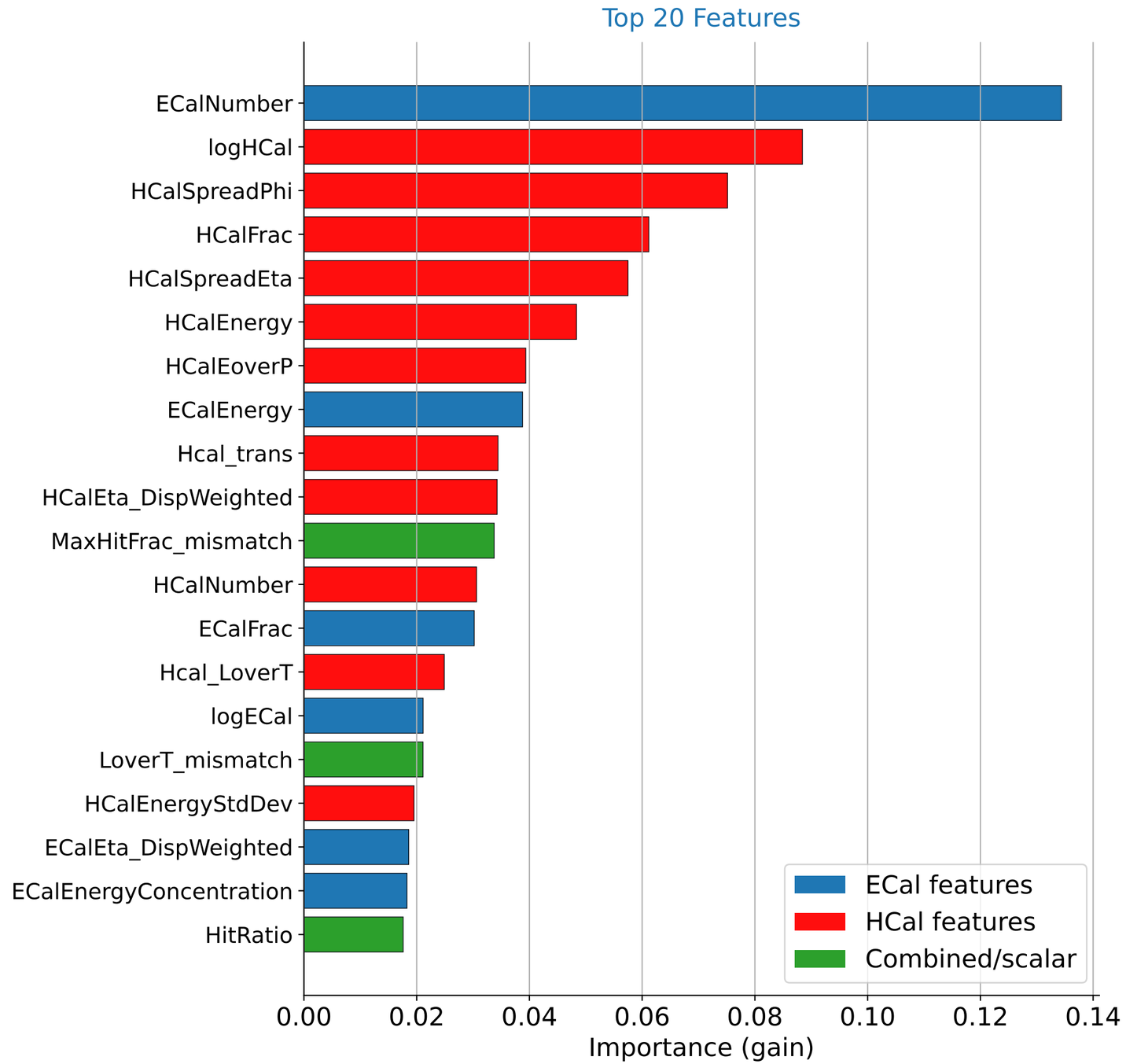


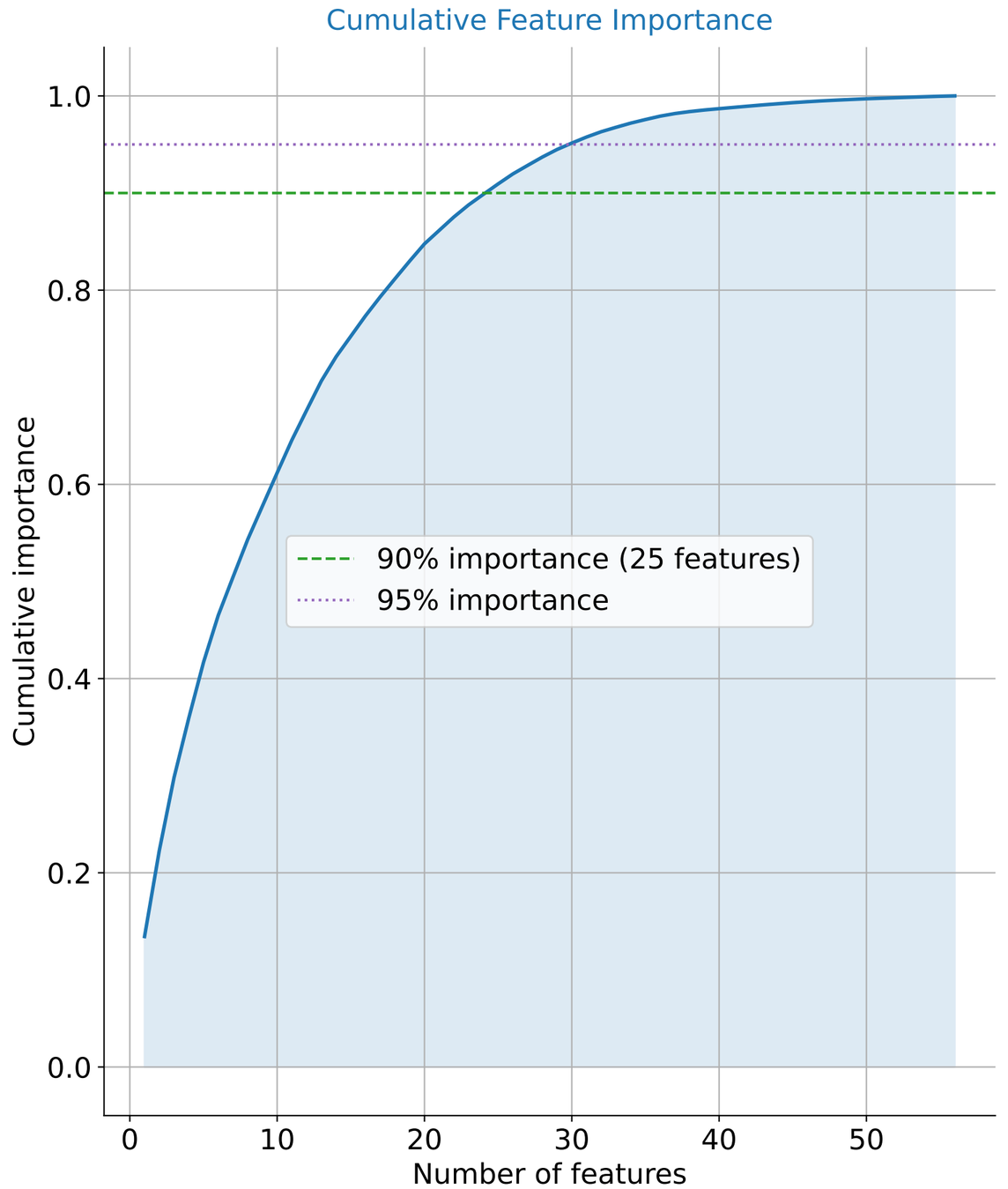
XGBoost Training Diagnostics



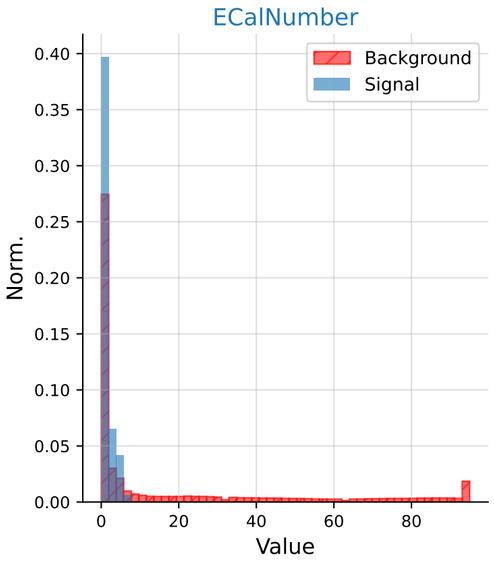


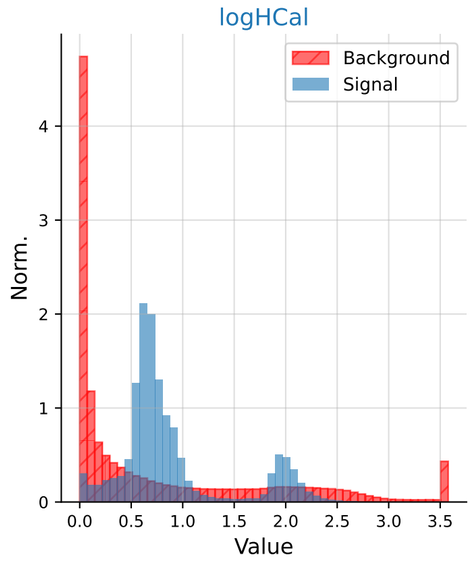
Feature Importances — XGBoost

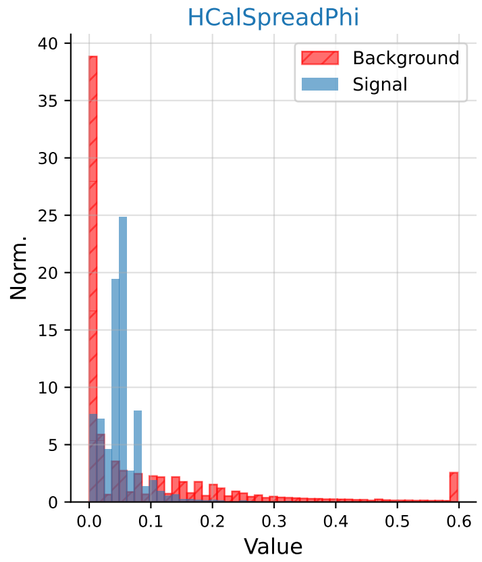


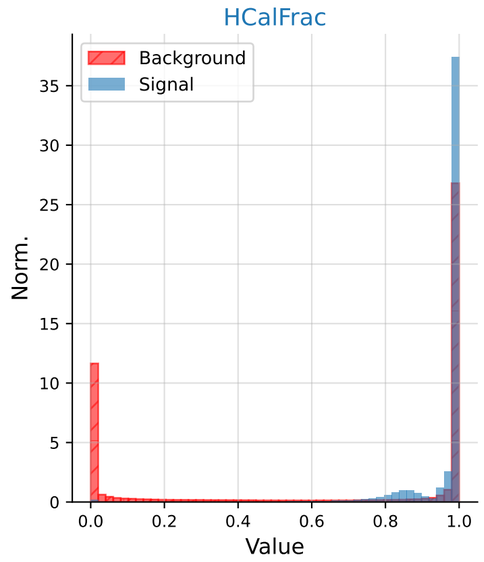


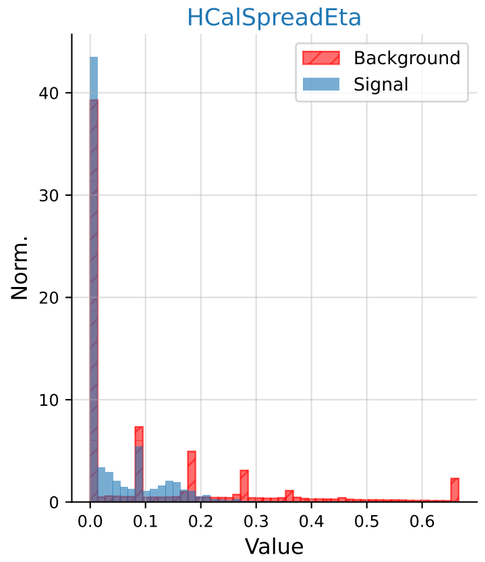
Key Physics Variable Distributions: Signal vs Background
[full dataset before splitting]

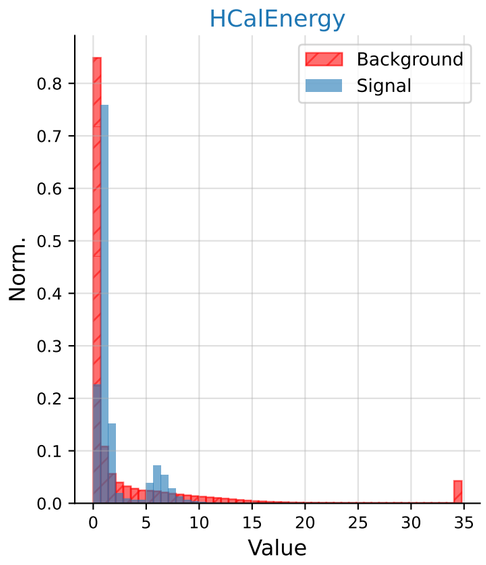


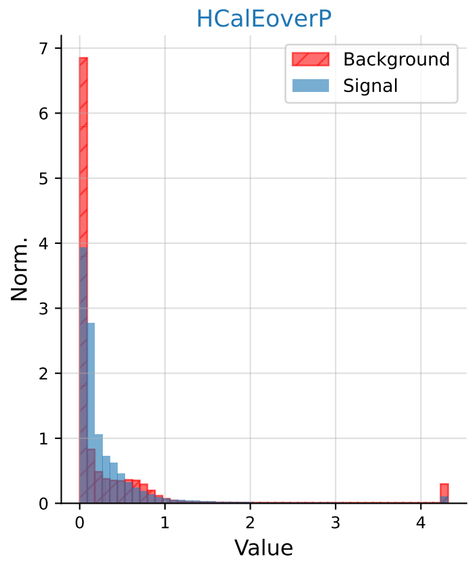


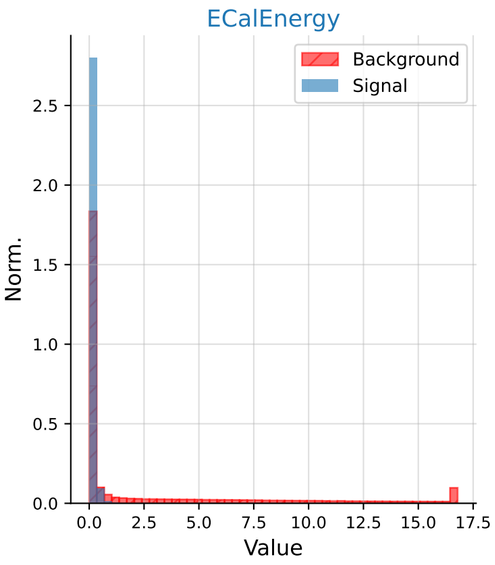


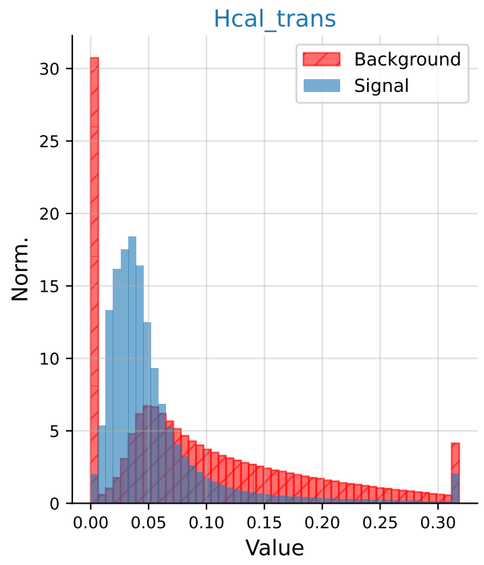


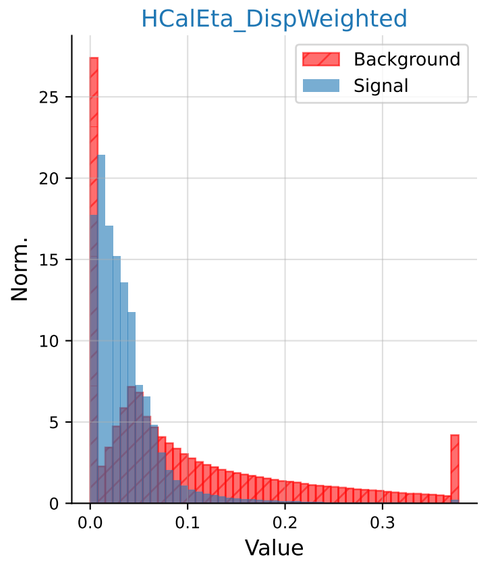


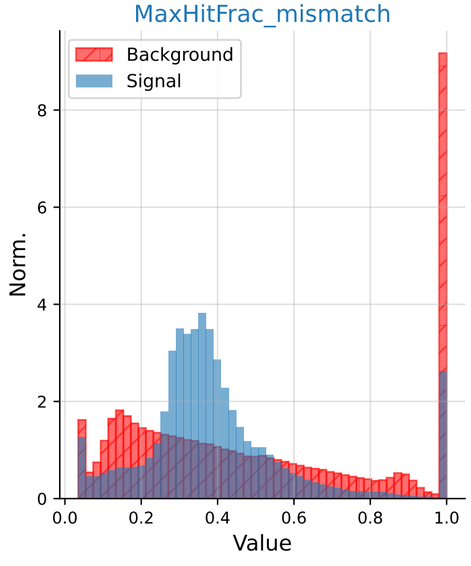


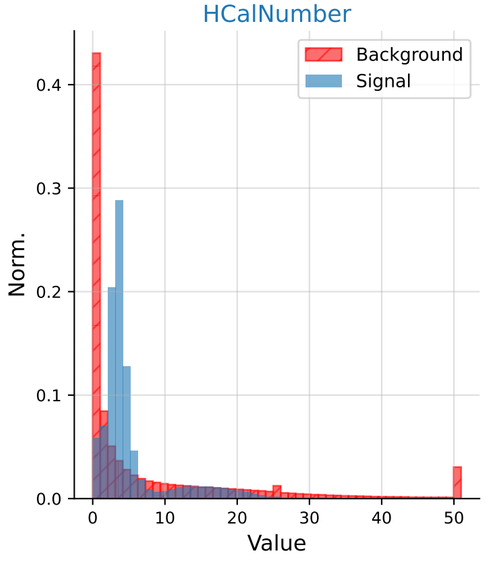


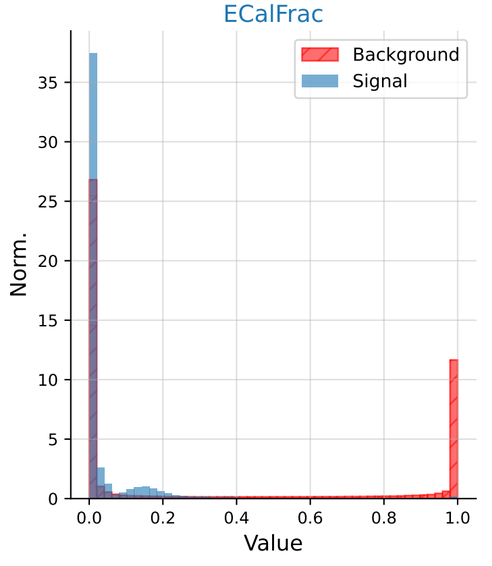


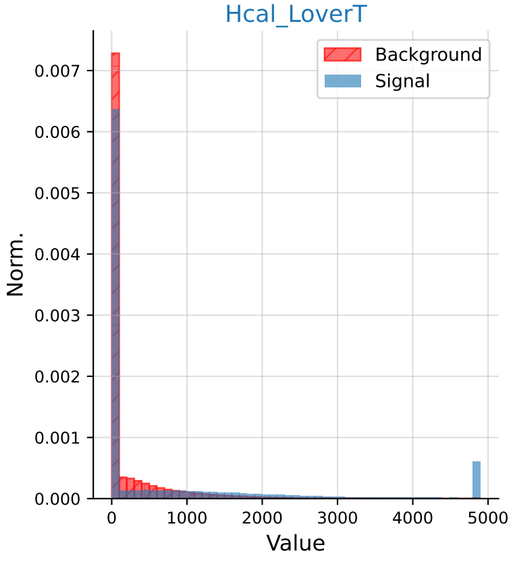


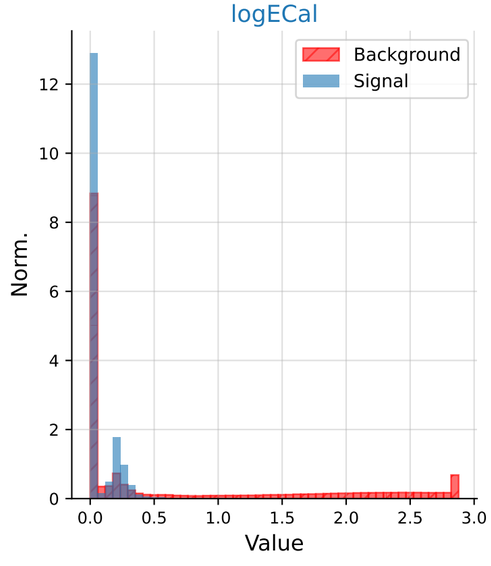






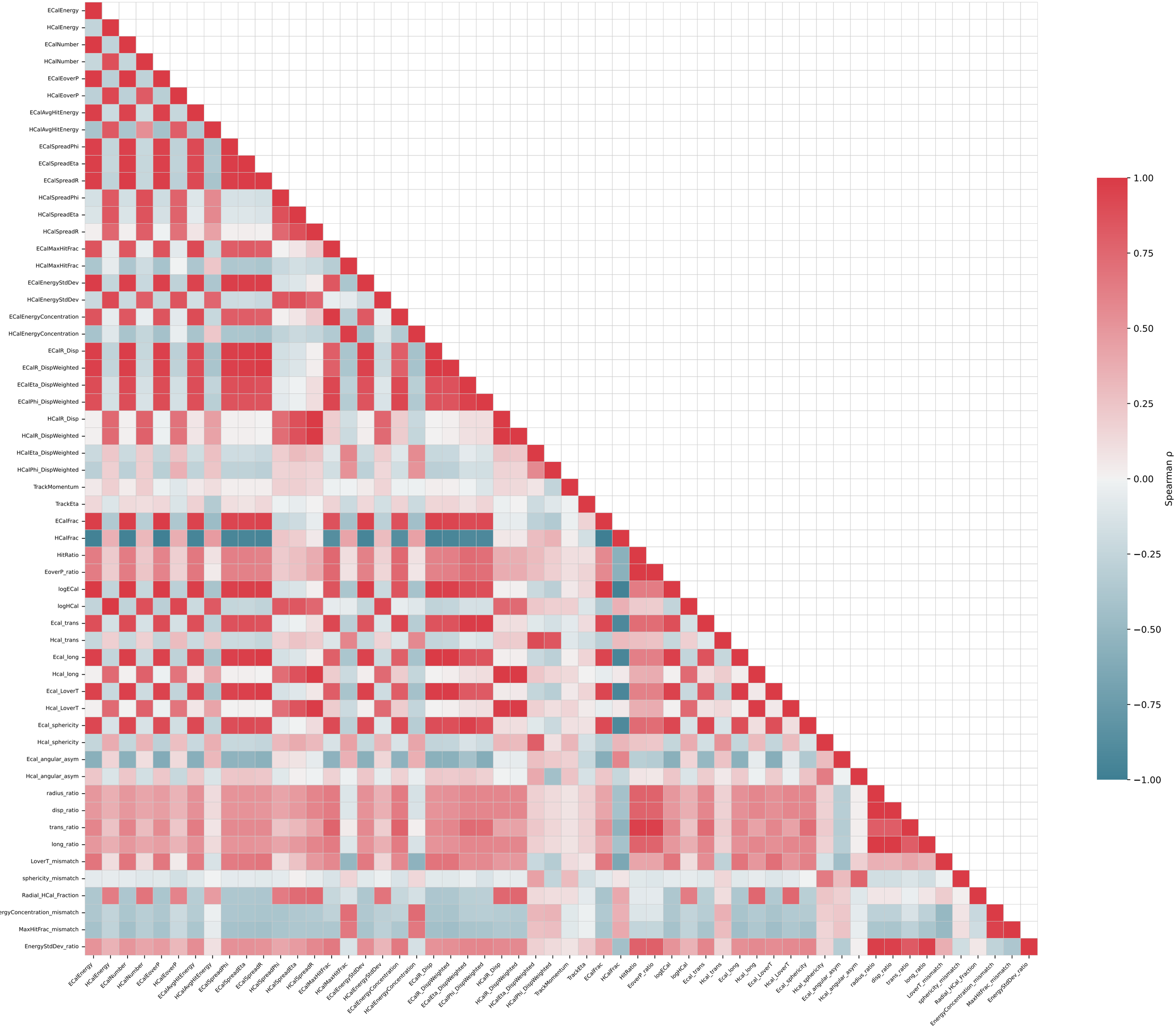






Spearman Correlation Matrix — All Features
(values from the full dataset)

Higher |ρ| means a stronger monotonic relationship between features



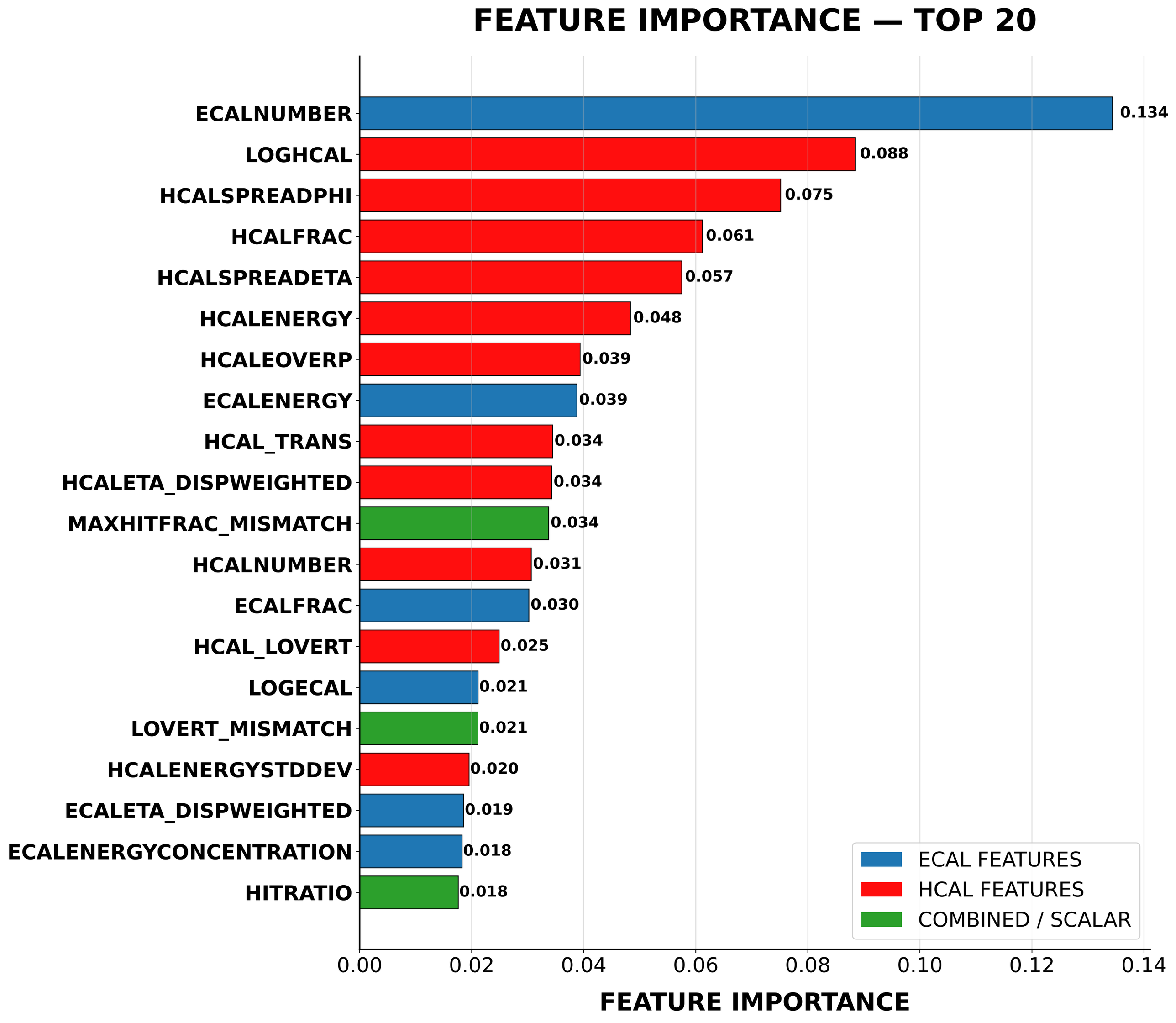
Feature correlation with signal/background class
Blue = positive (→ muon), Red = negative (→ background)

HCalEnergyConcentration
long_ratio
disp_ratio
radius_ratio
TrackEta
EnergyStdDev_ratio
trans_ratio
HCalPhi_DispWeighted
EnergyConcentration_mismatch
TrackMomentum
HCalSpreadEta
Ecal_angular_asym
HCalMaxHitFrac
HitRatio
EoverP_ratio
MaxHitFrac_mismatch
HCalSpreadR
HCalR_Disp
LoverT_mismatch
Hcal_long
HCalR_DispWeighted
HCalSpreadPhi
Hcal_sphericity
Hcal_LoverT
ECalEnergyConcentration
ECalMaxHitFrac
HCalNumber
HCalEnergyStdDev
Radial_HCal_Fraction
Hcal_trans
ECalPhi_DispWeighted
Ecal_LoverT
Ecal_trans
ECalAvgHitEnergy
Ecal_sphericity
HCalEnergy
logHCal
ECalEta_DispWeighted
Ecal_long
ECalR_DispWeighted
HCalEoverP
HCalEta_DispWeighted
ECalR_Disp
ECalSpreadEta
ECalEnergyStdDev
ECalSpreadR
ECalEoverP
ECalSpreadPhi
HCalFrac
ECalFrac
logECal
ECalEnergy
ECalNumber
sphericity_mismatch
HCalAvgHitEnergy
Hcal_angular_asym

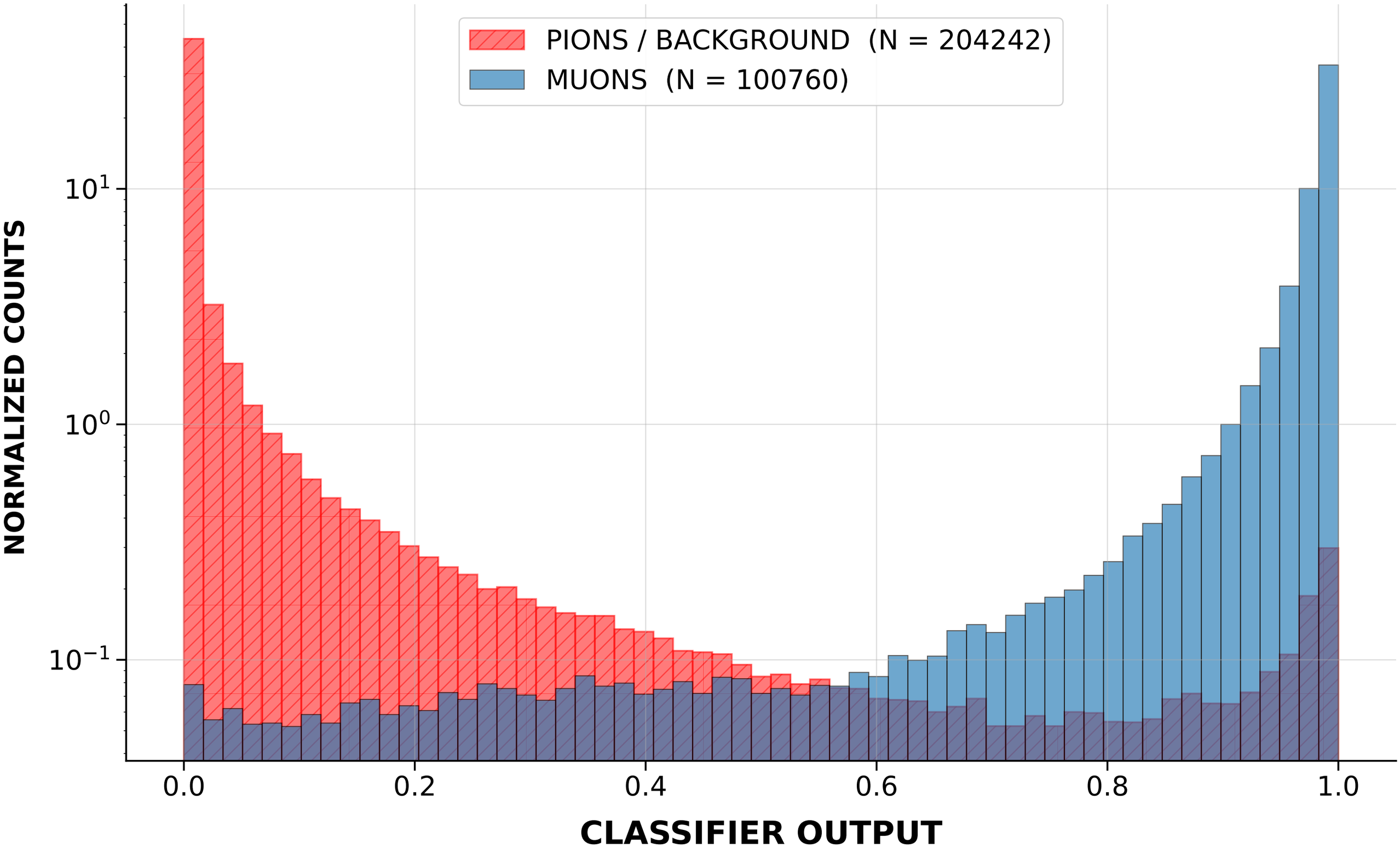
−0.4 −0.3 −0.2 −0.1 0.0 0.1 0.2 0.3

Spearman ρ with the label (IsMuon)

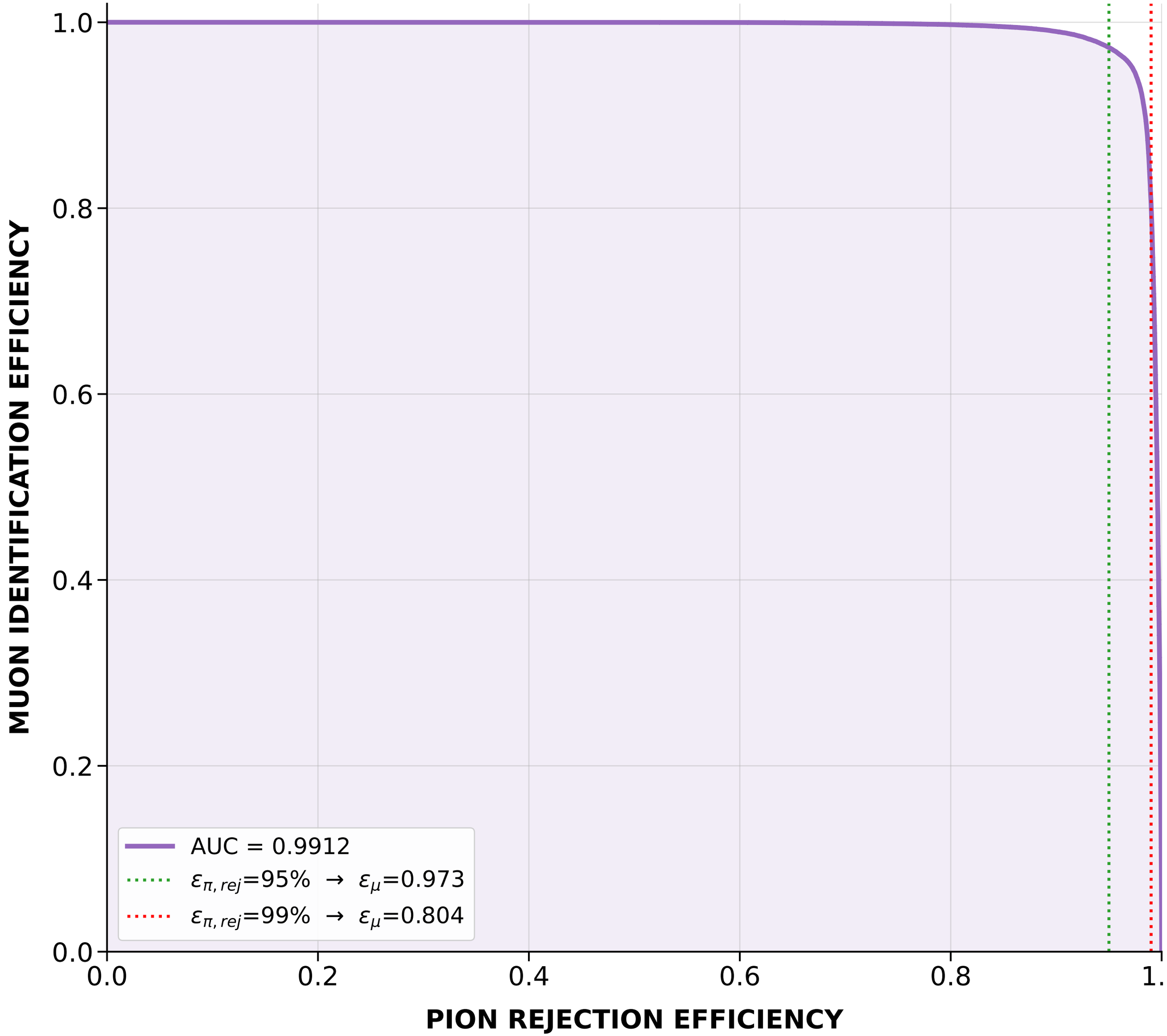
FEATURE IMPORTANCE — TOP 20



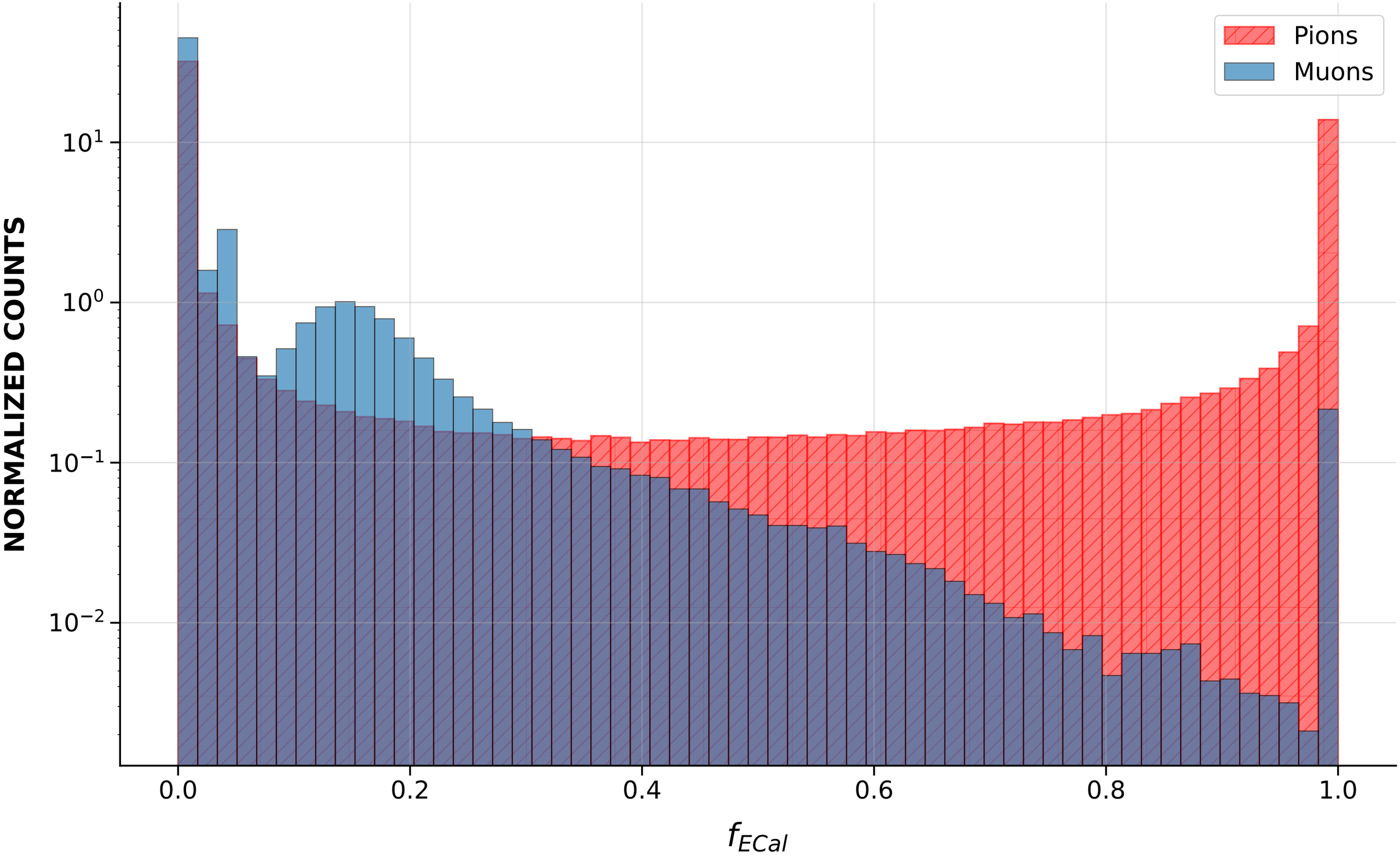
CLASSIFIER SCORE DISTRIBUTION
MUONS vs PIONS



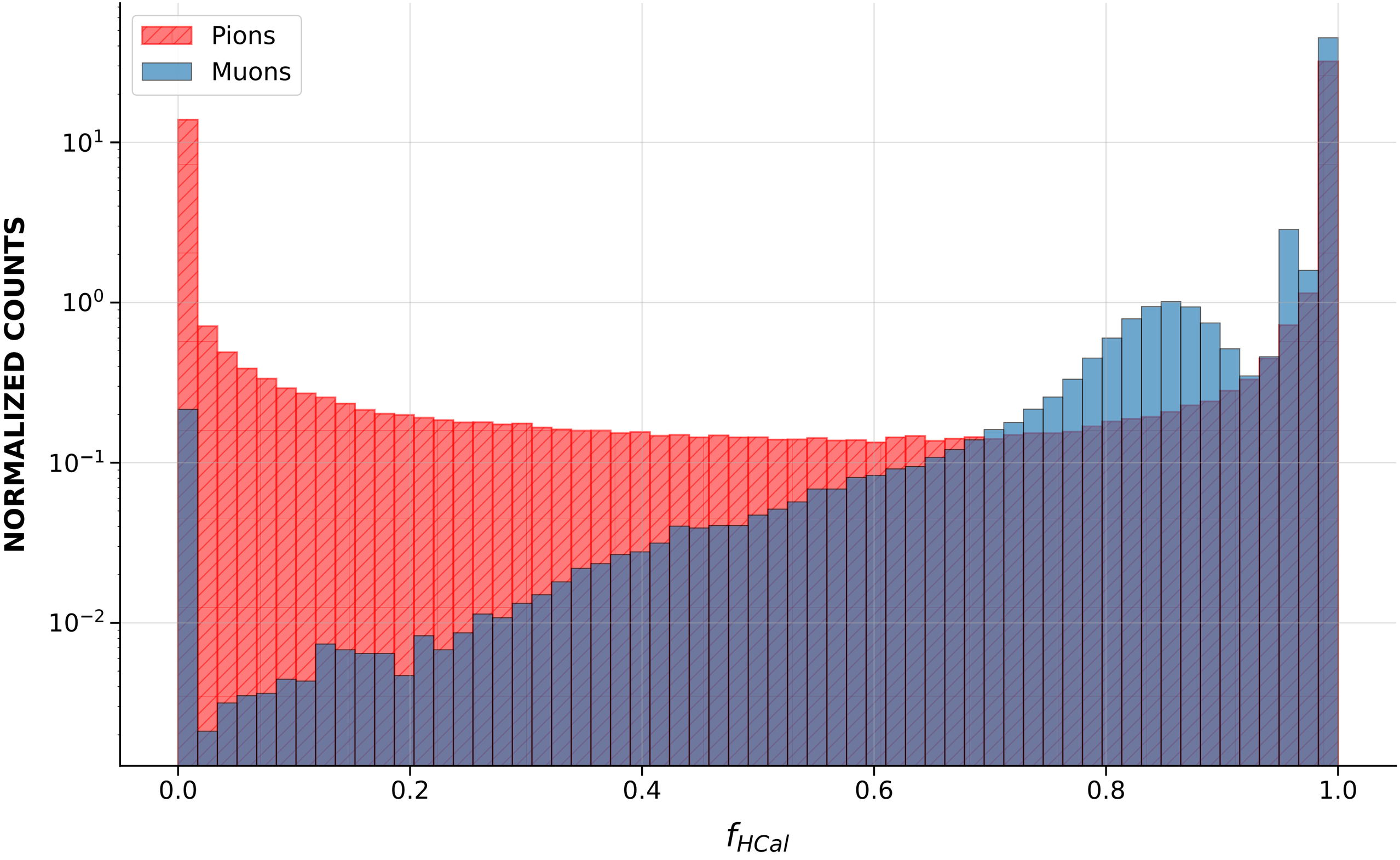
ROC CURVE: ε_μ vs $\varepsilon_{\pi,\text{rej}}$



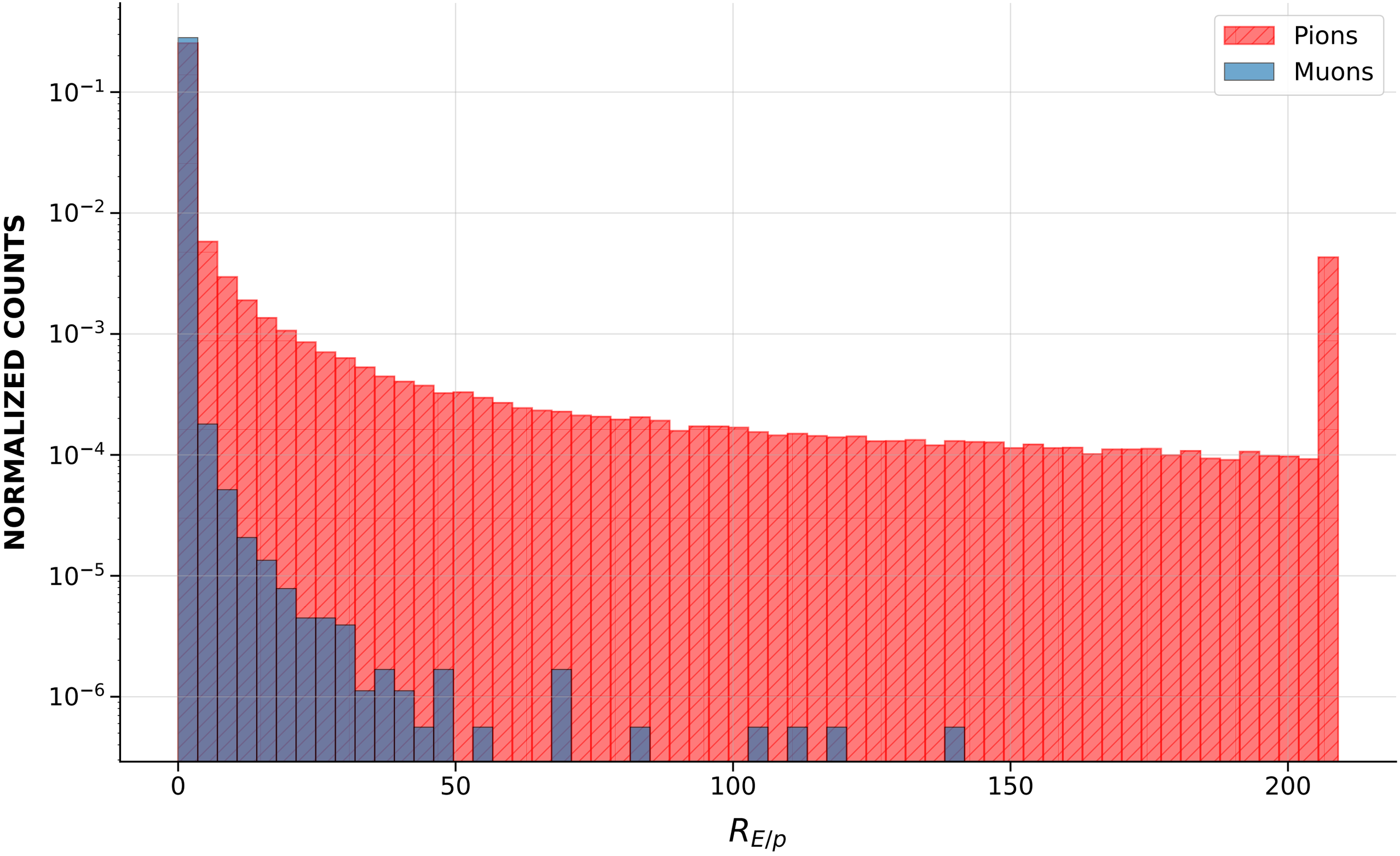
ENGINEERED CALORIMETER FEATURE
ECalFrac



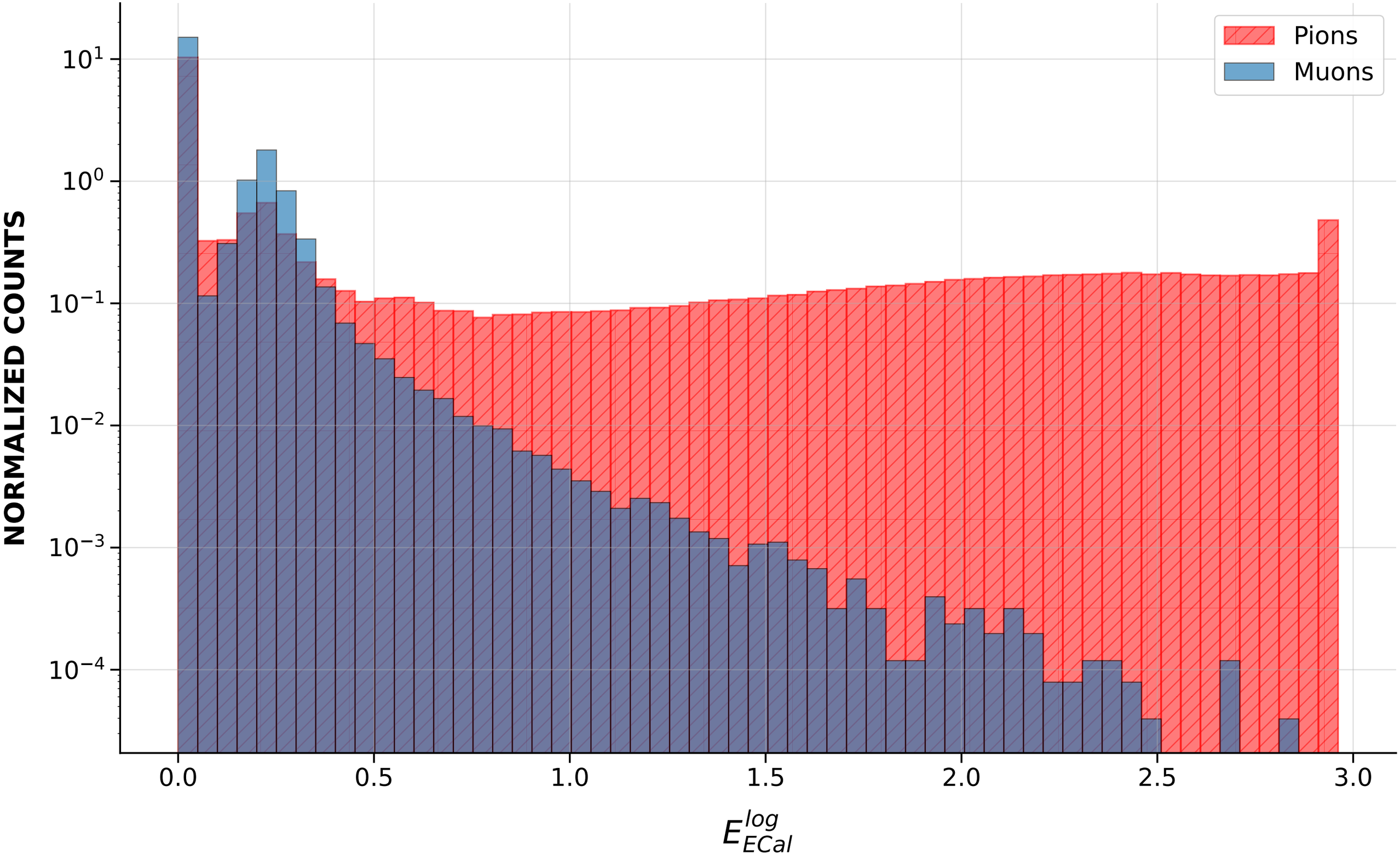
ENGINEERED CALORIMETER FEATURE
HCalFrac



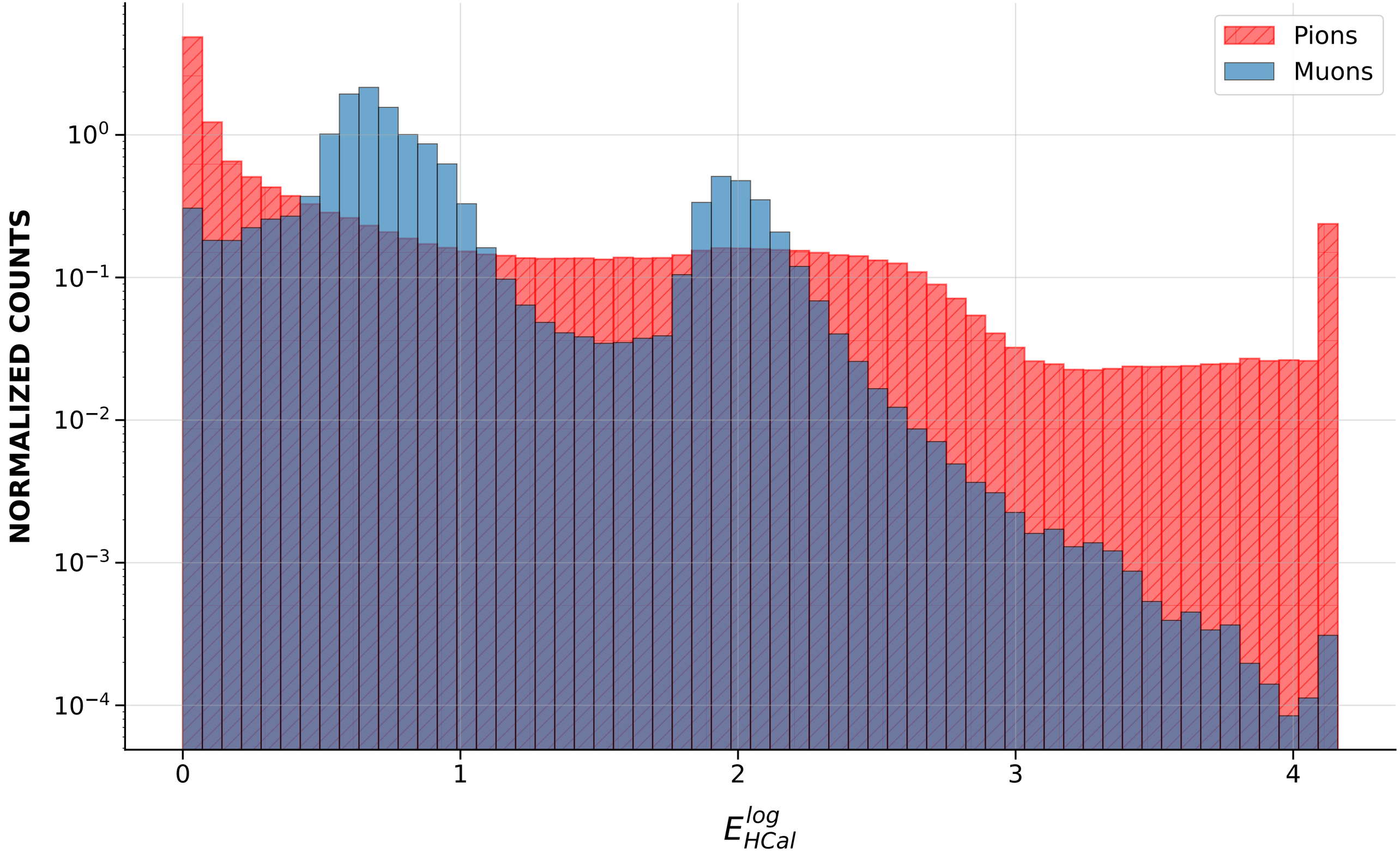
ENGINEERED CALORIMETER FEATURE
EoverP_ratio



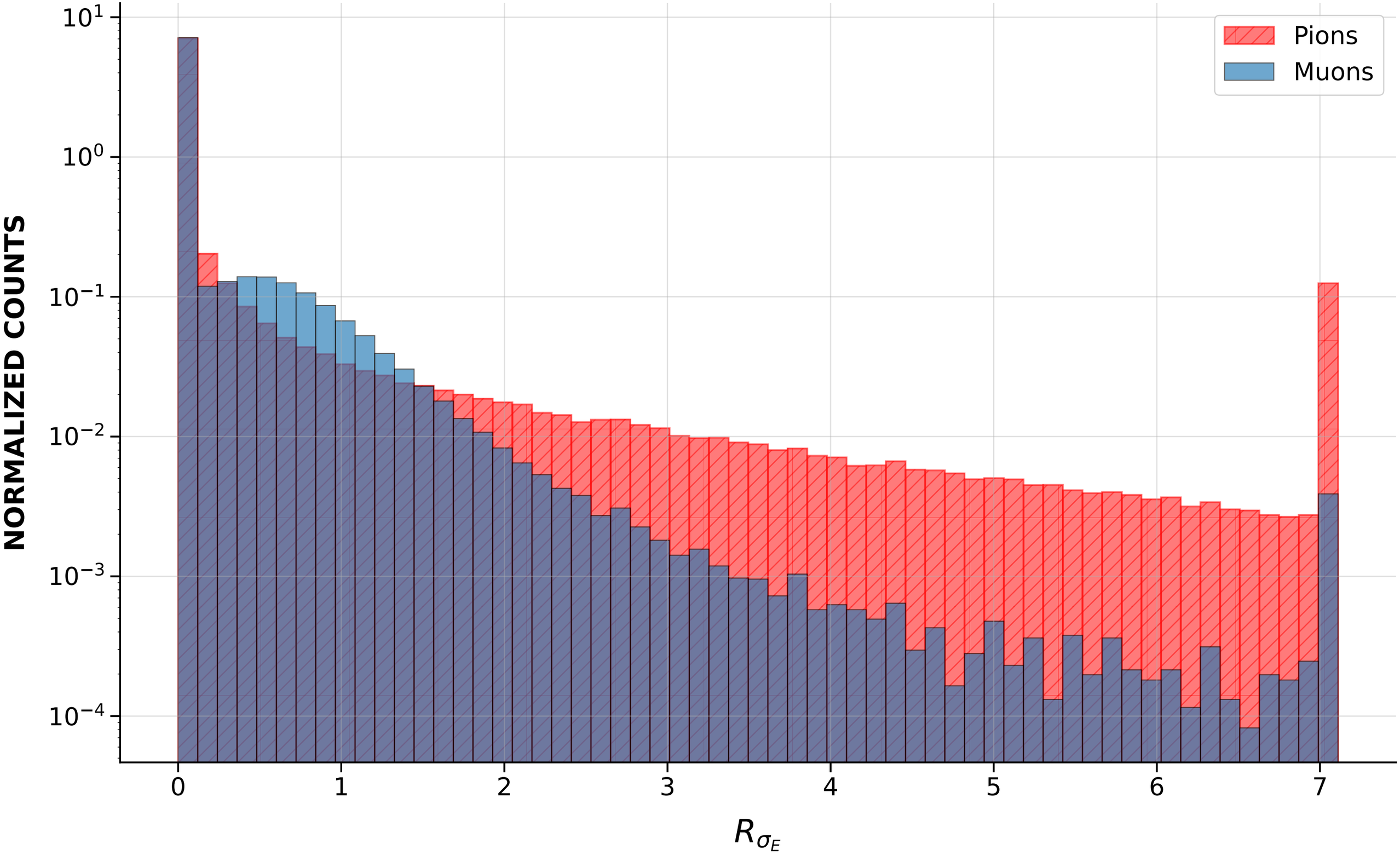
ENGINEERED CALORIMETER FEATURE
logECal



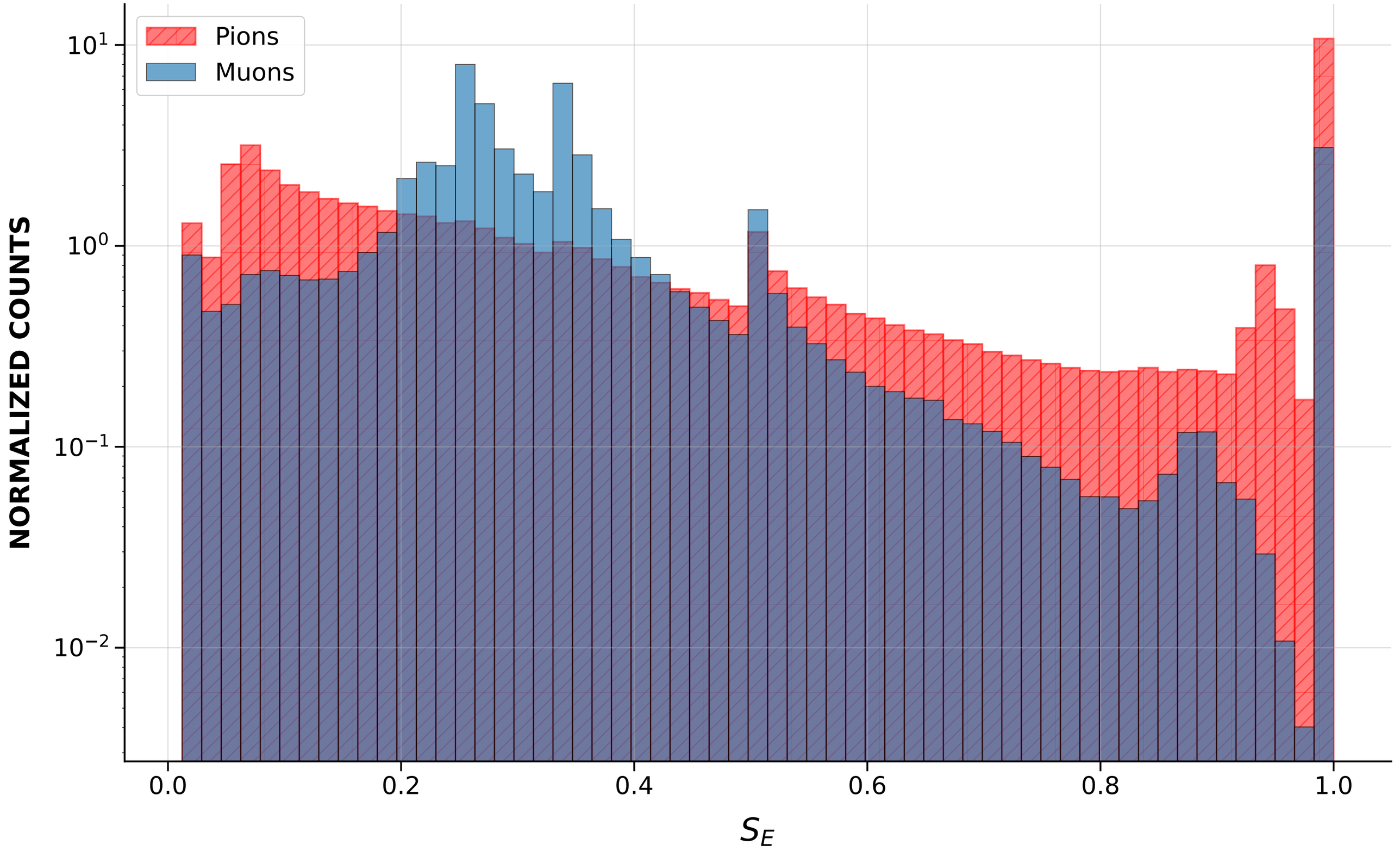
ENGINEERED CALORIMETER FEATURE
logHCal



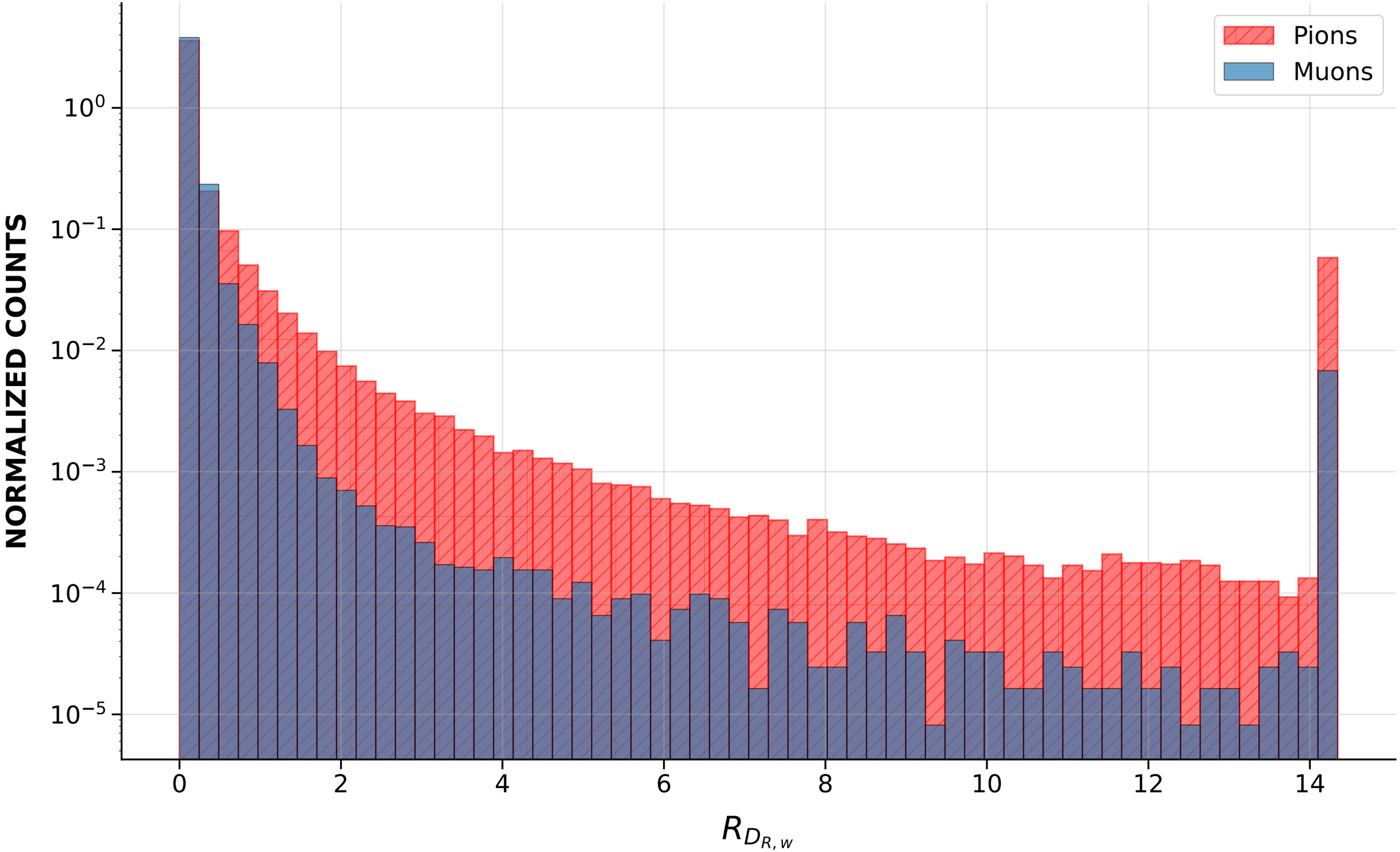
ENGINEERED CALORIMETER FEATURE
EnergyStdDev_ratio



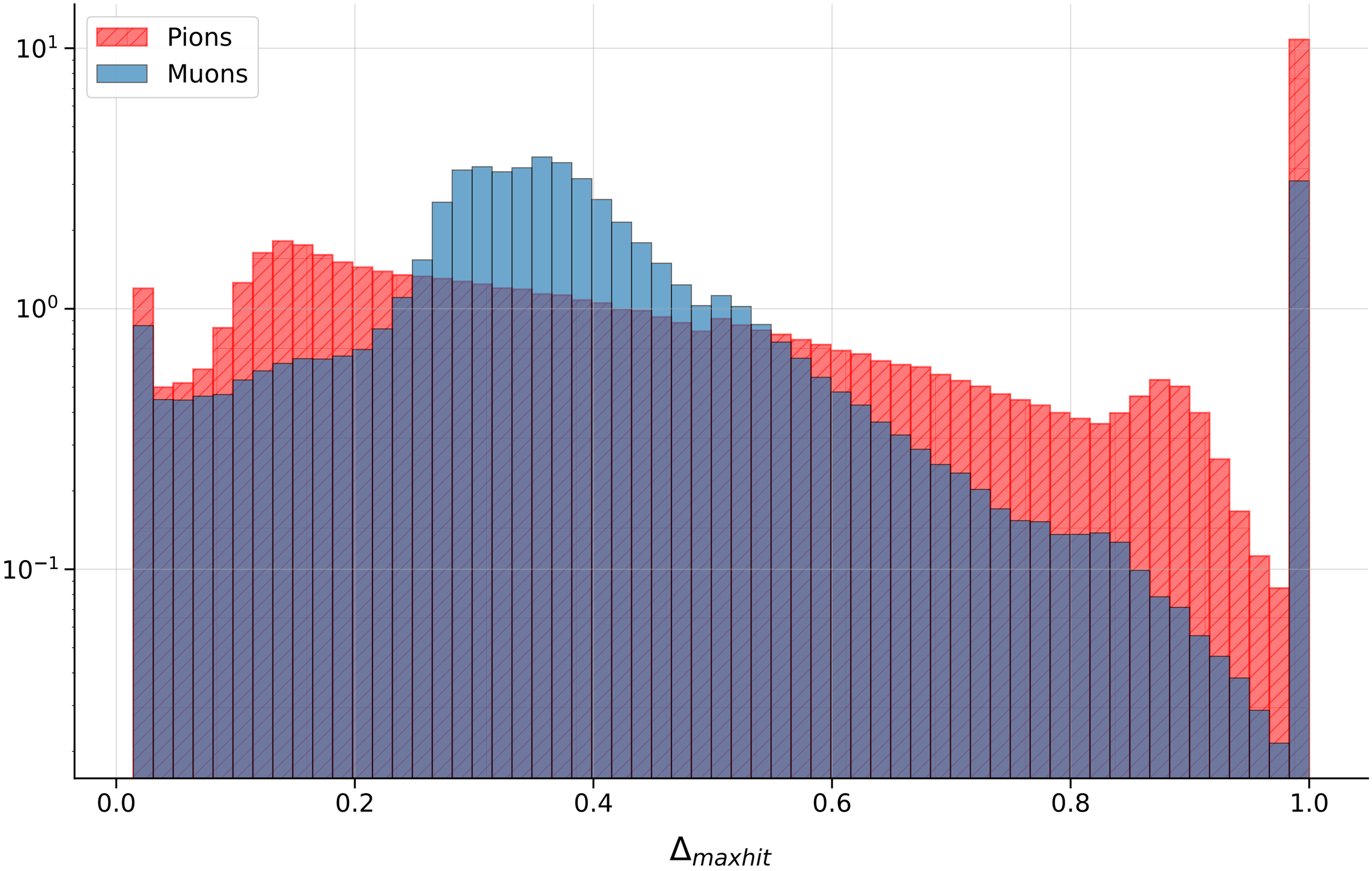
ENGINEERED CALORIMETER FEATURE
EnergyConcentration_mismatch



ENGINEERED CALORIMETER FEATURE
disp_ratio



ENGINEERED CALORIMETER FEATURE
MaxHitFrac_mismatch



ENGINEERED CALORIMETER FEATURE
HitRatio

